

Leakage-Resistant Rice Leaf Disease Classification Under Class Imbalance

An Empirical Study of Moderated Class Weighting and Cross-Split Generalization

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Artificial Intelligence • Deep Learning • Agricultural Computer Vision

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Core finding. I was excited to see that moderated class weighting had the validation results of the three methods tested (64.61 percent accuracy 58.41 percent Macro-F1, 63.22 percent Weighted-F1). On the reserved test set the chosen model scored 60.12 percent accuracy and 53.14 percent Macro-F1 showing a gap, between validation and test performance. I felt that the gap was meaningful.

Keywords: rice disease classification; MobileNetV2; class imbalance; class weighting; data leakage; duplicate auditing; generalization; agricultural AI; computer vision

Abstract

Automated rice disease classification with deep learning has produced strong results on a range of benchmark datasets, but reported performance can be difficult to interpret when datasets differ in composition, class balance, acquisition conditions, and evaluation protocols. This study investigates these issues using a leakage-resistant evaluation pipeline on the RiceLeafDiseaseBD Version 2 dataset. The original 9,769-image dataset was subjected to an exact-content audit that identified 39 duplicate groups involving 78 files, including eight duplicate groups with conflicting labels. After removing conflicting copies and redundant same-class copies, 9,722 unique images remained. The cleaned dataset was stratified into 7,777 training, 972 validation, and 973 locked held-out test images. A frozen ImageNet-pretrained MobileNetV2 classifier was evaluated under three controlled training strategies: an unweighted baseline, full inverse-frequency class weighting, and moderated class weighting using the square root of the inverse-frequency weights. Macro-F1 was used as the way to choose the best model. The test set was not used until the end. Using a way to weight things gave the best results during testing. This approach made the accuracy better by 1.65 percentage points and Macro-F1 better by 3.03 percentage points compared to the version.. The model that was chosen only had 60.12% accuracy and 53.14% Macro-F1 when tested on the test set. The class called Leaf Smut was the one to get right with only 11.54% test F1. Other classes, like Sheath Blight, Rice Tungro and Healthy had better F1 scores. Bootstrap confidence intervals further quantify uncertainty around the final test estimates. The results indicate that weighting strength matters, but also that balanced validation performance does not guarantee equally strong unseen-sample performance. The study therefore emphasizes dataset integrity, class-wise evaluation, held-out testing, and cautious interpretation of agricultural AI systems.

Practical interpretation. The selected model should be viewed as an empirical research baseline and not as a production-ready autonomous plant-disease diagnostic system, particularly because Leaf Smut recognition remains weak.

1. Introduction

Rice is a staple crop. When disease outbreaks happen, rice productivity. The cost and difficulty of managing the crop go up. Computer vision and deep learning can help farmers. By learning patterns that show whether leaves are healthy or diseased these technologies give useful decision support. Field-based datasets are particularly important because real agricultural imagery contains variation in illumination, backgrounds, disease severity, leaf orientation, and image quality. A systematic review of rice leaf disease research identified substantial methodological diversity and emphasized the need for more robust and reliable evaluation protocols [1].

A central challenge is that high reported accuracy can conceal weaknesses in minority classes or optimistic estimates caused by data overlap. Duplicate or near-duplicate images can cross partition boundaries and make

held-out evaluation less independent. Prior work on CIFAR demonstrated that duplicate contamination can materially affect measured generalization [2]. In agricultural datasets, where multiple images may originate from the same plant, scene, or acquisition session, this concern is especially relevant.

This study therefore focuses less on proposing a new neural architecture and more on evaluating the reliability of a lightweight classification pipeline. MobileNetV2 was selected because its inverted residual structure and depthwise convolutions provide a computationally efficient backbone suitable for resource-constrained applications [3]. The study compares unweighted learning with two levels of class weighting and evaluates the selected model on a locked held-out test split.

The dataset used here is RiceLeafDiseaseBD Version 2, a Bangladesh field-image dataset containing six classes: Blast, Brown Spot, Healthy, Leaf Smut, Rice Tungro, and Sheath Blight [4]. The dataset is not a Nepalese field dataset; Nepal is treated only as the broader application context. Consequently, the present experiments do not establish performance on Nepalese rice fields.

2. Related Work

2.1 Rice disease image classification

Paddy Doctor is a field-oriented resource that contains 16,225 images across 13 classes. In its benchmark experiments Paddy Doctor reported a ResNet34 F1-score of 97.50 percent. Recent work keeps exploring lightweight architectures, attention mechanisms, transfer learning and models designed for deployment. For example, Xu et al. developed LiSA-MobileNetV2 and reported high performance on a different Paddy Doctor-derived setup while substantially reducing model size [6]. Such results demonstrate the potential of lightweight models, but they also illustrate why numerical comparison across studies must account for dataset and protocol differences.

2.2 Dataset reliability and generalization

Recent evidence syntheses emphasize dataset provenance, split design, class imbalance, external validation, and deployment readiness as important dimensions of rice disease AI evaluation [7]. Dataset-specific studies also keep adding field collections, like BanglaRiceLeaf, which holds 4,152 pictures in five categories collected in real field settings, in Bangladesh. These datasets are useful. Because each set uses different class names and different ways of taking pictures it is not good to compare them directly on a leaderboard.

2.3 Class imbalance

Class imbalance can cause a classifier to optimize aggregate performance while under-serving minority categories. Class weighting addresses this by increasing the loss contribution of underrepresented classes. However, strong weighting can shift the decision boundary too aggressively, potentially improving minority recall at the expense of other classes. This motivates an empirical comparison between full inverse-frequency weighting and a moderated alternative.

3. Research Gap and Objectives

Existing rice disease studies vary substantially in dataset composition, class balance, acquisition conditions, architecture, and evaluation methodology. The present study addresses a narrower methodological gap: controlled evaluation of a lightweight classifier after explicit exact-duplicate auditing, with a systematic comparison of weighting strength and a genuinely locked within-dataset test set.

The goals are: (1) check the data set for duplicates and labels that disagree with each other; (2) create a clean split that can be reproduced; (3) test training of MobileNetV2, with no weights, full weights and some weights; (4) pick a model based on validation Macro-F1 without looking at the test set; and (5) measure how well the model performs on the test data how sure it is and how well it works for each class.

Research questions. RQ1: How does exact-duplicate and conflicting-label auditing affect dataset reliability?

RQ2: How does class weighting affect MobileNetV2 performance, on the six-class dataset that we are working with?

RQ3: Can moderated weighting give a balance than full inverse-frequency weighting?

RQ4: How well does the selected model generalize to a locked held-out test set and which classes show the largest weaknesses that we observe?

4. Materials and Methods

4.1 Dataset

RiceLeafDiseaseBD Version 2 has 9,769 high-resolution images. Each image is 1024×1024 pixels. Was taken with a smartphone. The pictures were collected in rice fields, in Gazipur, Bangladesh. The collection happened during the Summer 2025 growing season. The dataset covers six classes and includes field backgrounds and natural acquisition conditions [4]. The study used the original-image classification folders rather than the dataset's object-detection annotations.

Class	Original	Clean	Clean %
Blast	1,326	1,323	13.61%
Brown Spot	2,178	2,176	22.38%
Healthy	1,575	1,573	16.18%
Leaf Smut	724	721	7.42%
Rice Tungro	2,244	2,238	23.02%
Sheath Blight	1,722	1,691	17.39%
Total	9,769	9,722	100%

Table 1. Original and cleaned class distribution.
Class Distribution of the Clean Rice Leaf Disease Dataset

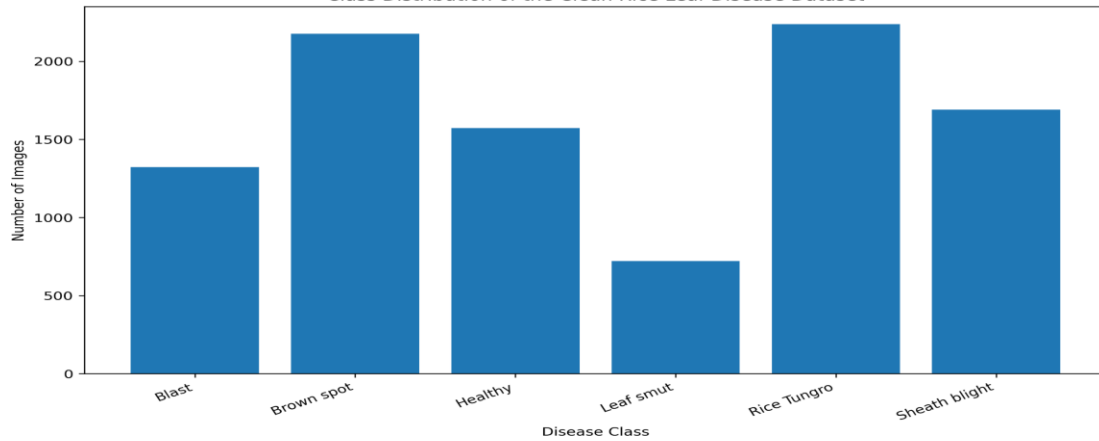


Figure 1. Class distribution after the exact-content cleaning procedure. Leaf Smut is the smallest class, while Rice Tungro and Brown Spot are the largest.

4.2 Data quality audit

All 9,769 images were audited using exact-content hashing. The audit found 9,730 unique image contents, 39 duplicate hash groups, and 78 files involved in duplicate groups. Eight groups contained identical image content assigned to different class labels. Because an exact duplicate cannot legitimately provide two independent visual observations, conflicting-label copies were removed rather than choosing a label based on visual judgment. Same-class duplicate groups were reduced to one representative. In total, 47 files were removed: 16 from conflicting-label groups and 31 redundant same-class copies. The cleaned dataset contained 9,722 images, with zero remaining exact duplicate groups and zero remaining conflicting-label groups.

This procedure addresses exact duplication but does not prove that all near-duplicates or correlated acquisition events have been eliminated. Near-duplicate leakage is a broader problem documented in computer-vision benchmarks [2].

4.3 Dataset splitting

The cleaned image list was split using stratified random sampling with seed 42. The final split contained 7,777 training images, 972 validation images, and 973 locked held-out test images. The test set was locked before model selection and was not used to tune training strategy, hyperparameters, or weighting strength.

Subset	Images	Role
Training	7,777	Model fitting
Validation	972	Model selection / training monitoring
Held-out test	973	Final evaluation only
Total	9,722	

Table 2. Final leakage-resistant dataset split.

4.4 Image preprocessing and augmentation

Images were resized to 224×224 pixels. Training images were augmented using random horizontal flipping, rotation up to approximately 15°, zoom up to approximately 15%, and contrast adjustment up to approximately 15%.

Validation and test images were not randomly augmented. MobileNetV2 preprocessing was applied before feature extraction.

4.5 Model architecture

I used a classifier that had an ImageNet-pretrained MobileNetV2 backbone with the convolutional backbone frozen. The classification head was Input → augmentation → MobileNetV2 preprocessing → MobileNetV2 → average pooling → dropout (0.4) → dense 128 ReLU → dropout (0.3) → six-unit softmax. The model contained 2,422,726 parameters, of which 164,742 were trainable.

4.6 Experimental strategies

Three controlled strategies were evaluated with the same architecture, preprocessing, optimizer, batch size, and training budget.

Experiment	Training loss weighting	Purpose
1. Baseline	$w_c = 1$	Unweighted reference
2. Full weighting	$w_c = N / (K n_c)$	Inverse-frequency correction
3. Moderated weighting	$w'_c = \sqrt{w_c}$	Reduced weighting magnitude

Table 3. Controlled training strategies. The moderated strategy is the square root of the full inverse-frequency weight.

For full weighting, N is the number of training images, K is the number of classes, and n_c is the training count for class c . The moderated strategy takes the square root of the full inverse-frequency weight, reducing extreme emphasis on the smallest class.

4.6 Training protocol

All experiments used a batch size of 32; experiments used the Adam optimizer with a learning rate of 1×10^{-4} ; experiments used sparse categorical cross-entropy; experiments ran for a maximum of 12 epochs; experiments used a random seed of 42. Model checkpointing selected the lowest validation loss. Early stopping and ReduceLROnPlateau were used during training. Macro-F1 was designated as the primary model-selection metric, with accuracy and Weighted-F1 treated as secondary aggregate measures.

4.7 Evaluation and uncertainty

Performance was assessed using accuracy, macro-averaged precision, macro-averaged recall, Macro-F1, Weighted-F1, per-class precision, recall, F1-score, and confusion matrices. Five thousand bootstrap resamples were used to estimate 95% confidence intervals for the final test accuracy, Macro-F1, and Weighted-F1. Paired bootstrap comparisons were also performed between validation predictions from the three strategies.

5. Results

5.1 Validation comparison

Model	Accuracy	Macro-F1	Weighted-F1
Baseline	62.96%	55.38%	61.01%
Full Class Weighting	60.49%	56.87%	60.41%
Moderated Class Weighting	64.61%	58.41%	63.22%

Table 4. Validation performance of the three training strategies.

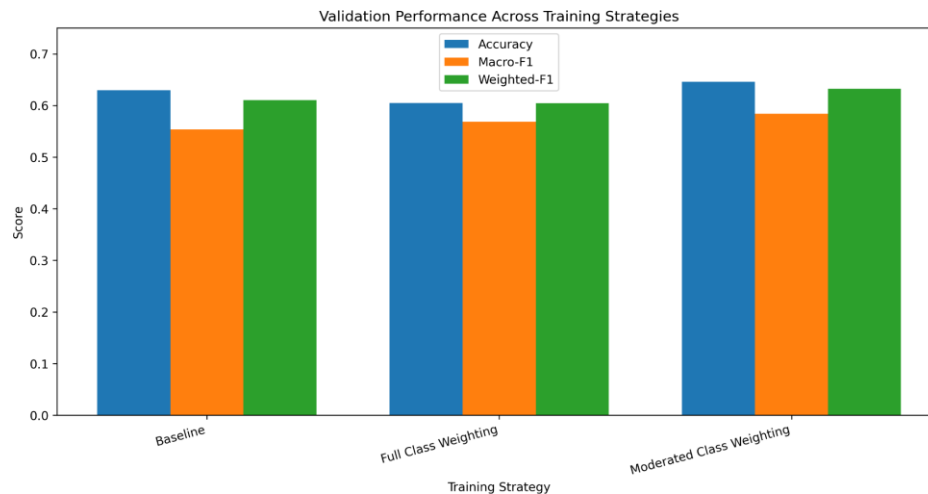


Figure 2. Validation performance across the three controlled training strategies.

Moderated weighting produced the validation accuracy, Macro-F1 and Weighted-F1. Compared to the baseline Moderated weighting accuracy by 1.65 percentage points Macro-F1 by 3.03 percentage points and Weighted-F1, by 2.21 percentage points. Full weighting improved Macro-F1 over the baseline but reduced accuracy. Reduced Weighted-F1.

5.2 Per-class validation performance

Class	Baseline F1	Full weighting F1	Moderated F1
Blast	40.72%	44.08%	44.36%
Brown Spot	57.36%	44.95%	59.54%
Healthy	67.60%	69.01%	72.83%
Leaf Smut	8.43%	27.65%	22.22%
Rice Tungro	72.85%	71.30%	72.49%
Sheath Blight	72.02%	74.76%	79.01%

Table 5. Validation F1-score by disease class.

Full weighting substantially increased Leaf Smut F1 from 8.43% to 27.65%, but this was accompanied by a large Brown Spot decline from 57.36% to 44.95%. Moderated weighting recovered Brown Spot performance and achieved the highest F1 for Blast, Brown Spot, Healthy, and Sheath Blight among the evaluated strategies. These results suggest a trade-off rather than a universal benefit from stronger weighting.

5.3 Experiment 3 training dynamics

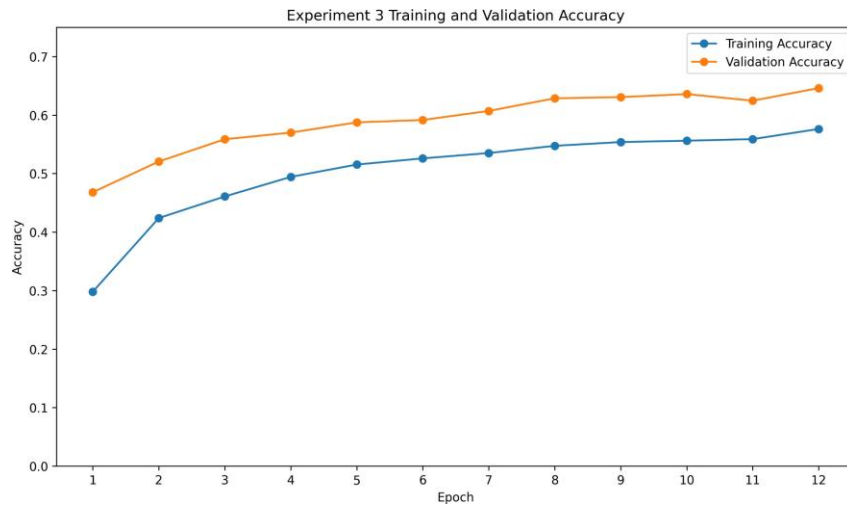


Figure 3. Training and validation accuracy for Experiment 3. Validation accuracy increased from 46.81% at epoch 1 to 64.61% at epoch 12.

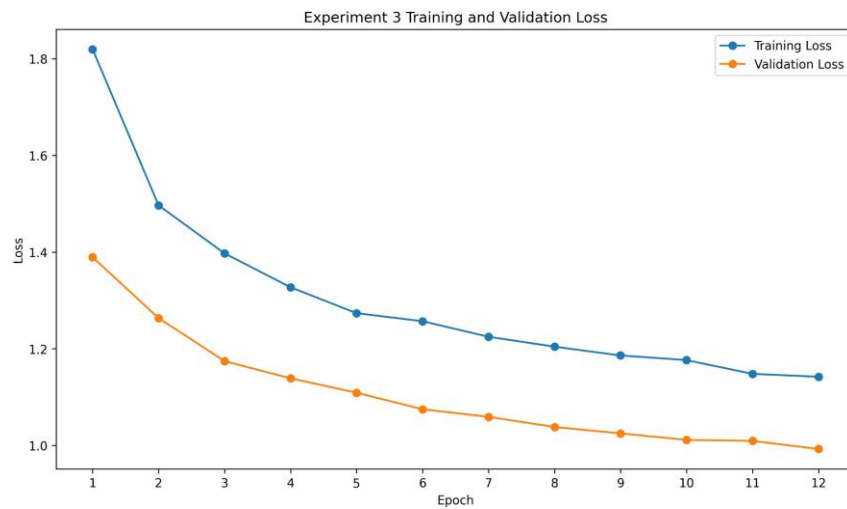


Figure 4. Training and validation loss for Experiment 3. Validation loss decreased from 1.3895 to 0.9926 over the 12-epoch run.

The selected Experiment 3 run showed steadily decreasing training and validation loss over the full 12 epochs. Validation accuracy also increased overall, reaching 64.61% at epoch 12. The higher validation accuracy than training accuracy should not be interpreted as evidence of data leakage: training was performed with augmentation and class-weighted loss, which changes the difficulty and optimization objective of training batches relative to unaugmented validation evaluation.

5.4 Held-out test performance

Metric	Held-out test
Accuracy	60.12%
Macro Precision	53.04%
Macro Recall	54.75%
Macro-F1	53.14%
Weighted-F1	58.24%

Table 6. Final held-out-test performance of the selected Experiment 3 model.

Class	Precision	Recall	F1
Blast	43.61%	43.94%	43.77%
Brown Spot	56.79%	42.20%	48.42%
Healthy	62.89%	77.22%	69.32%
Leaf Smut	18.75%	8.33%	11.54%
Rice Tungro	66.80%	76.34%	71.25%
Sheath Blight	69.39%	80.47%	74.52%

Table 7. Held-out-test performance by class.

The model that was chosen was tested on the hidden set of 973 images. The accuracy went down from 64.61% during validation to 60.12% during testing. At the time the Macro-F1 score dropped from 58.41% to 53.14%. The class called Sheath Blight had the F1 score, which was 74.52%. Next was Rice Tungro with 71.25% and Healthy with 69.32%. The class called Leaf Smut had the score. Only 6 out of 72 images, in the test set were classified correctly for Leaf Smut.

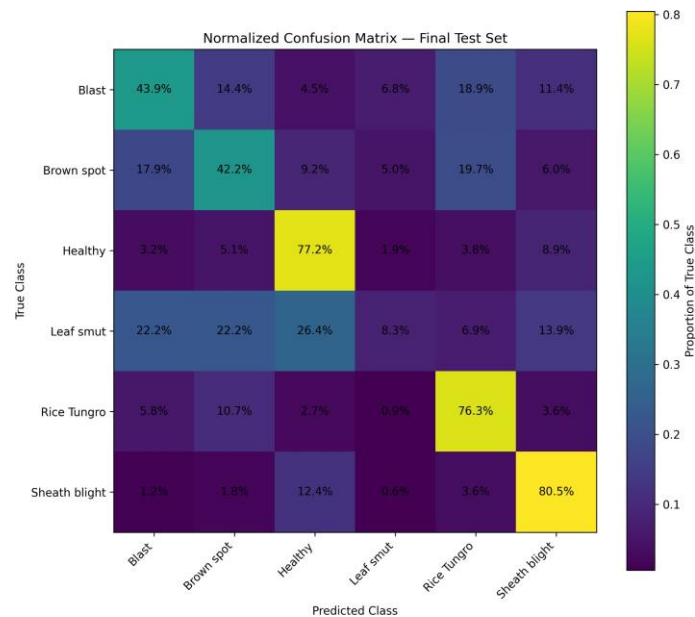


Figure 5. Normalized confusion matrix for the locked held-out test set. Diagonal values represent the proportion of each true class correctly classified.

5.5 Validation-to-test generalization

Class	Validation F1	Test F1	Drop
Blast	44.36%	43.77%	−0.59 pp
Brown Spot	59.54%	48.42%	−11.12 pp
Healthy	72.83%	69.32%	−3.51 pp
Leaf Smut	22.22%	11.54%	−10.68 pp
Rice Tungro	72.49%	71.25%	−1.24 pp
Sheath Blight	79.01%	74.52%	−4.49 pp

Table 8. Validation-to-test F1 generalization.

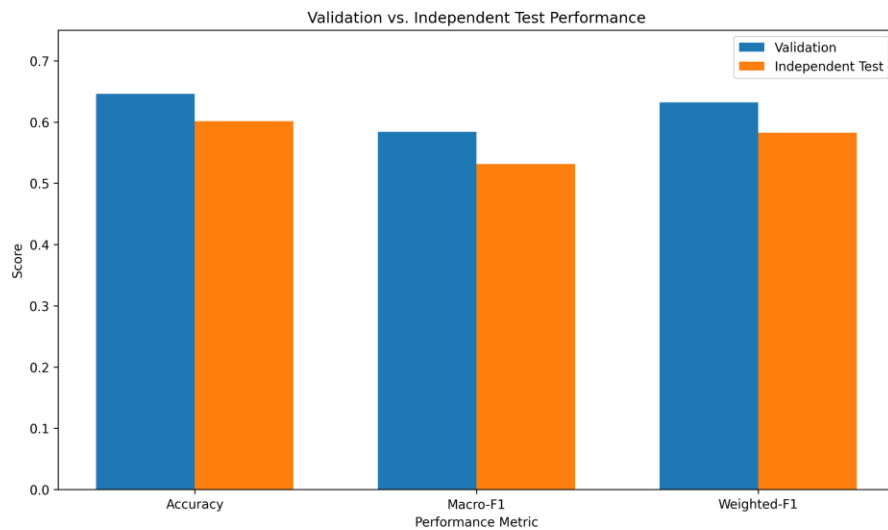


Figure 6. Aggregate validation versus held-out-test performance. The held-out test shows lower performance on all three aggregate metrics.

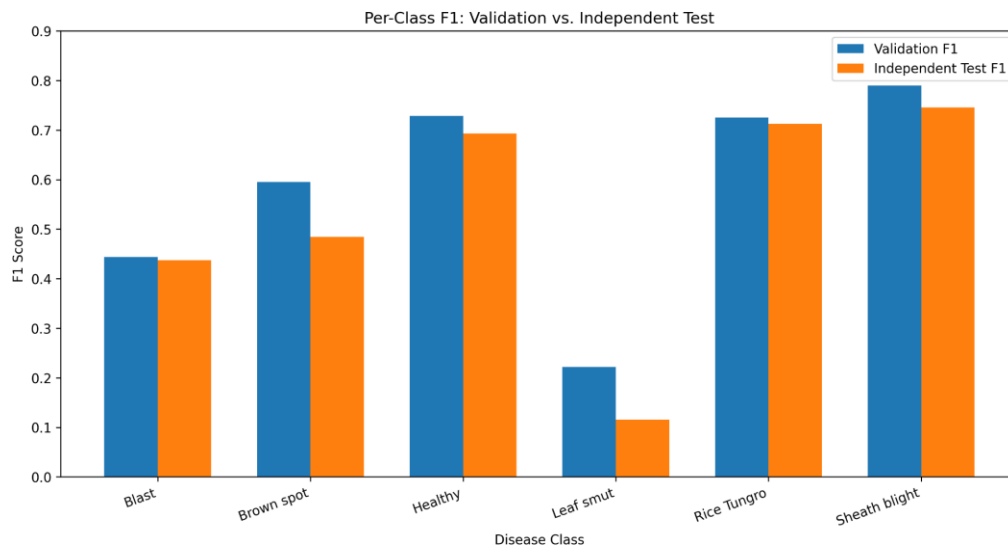


Figure 7. Per-class F1 on validation and held-out test sets. Brown Spot and Leaf Smut show the largest declines.

5.5 Validation-to-test generalization (continued)

Generalization degradation was class dependent. Brown Spot and Leaf Smut recorded the drops in F1 scores while Blast and Rice Tungro stayed more steady. The overall difference in Macro-F1 was five point twenty seven percentage points, with 58.41 percent compared to 53.14 percent. This demonstrates why validation performance alone can provide an optimistic estimate of unseen-sample performance.

5.6 Error analysis

The most frequent test-set confusion was Brown Spot → Rice Tungro (43 images), followed by Brown Spot → Blast (39). Other notable errors included Blast → Rice Tungro (25), Rice Tungro → Brown Spot (24), Sheath Blight → Healthy (21), Brown Spot → Healthy (20), Blast → Brown Spot (19), and Leaf Smut → Healthy (19). The normalized confusion matrix indicates that only 8.3% of Leaf Smut images were correctly recognized; 26.4% were classified as Healthy, 22.2% as Blast, and 22.2% as Brown Spot.

5.7 Bootstrap confidence intervals

Metric	Estimate	95% Bootstrap CI
Accuracy	60.12%	57.14–63.10%
Macro-F1	53.14%	50.17–56.05%
Weighted-F1	58.24%	55.02–61.46%

Table 9. Bootstrap confidence intervals for final test metrics (5,000 resamples, seed 42).

The paired validation bootstrap comparison estimated a Macro-F1 difference of percentage points for moderated weighting versus baseline with a 95 % bootstrap interval of +0.22 to +5.80 percentage points. The moderated-versus-full-weighting difference was +1.54 percentage points with a bootstrap interval of −1.15, to +4.16 percentage points, which includes zero. These bootstrap intervals are used as uncertainty summaries. Should not be interpreted as formal hypothesis-test p-values.

6. Discussion

6.1 Effect of weighting strength

The results support the view that weighting strength matters. Full inverse-frequency weighting improved Macro-F1 relative to the unweighted baseline, largely because minority-class recognition improved, but it reduced aggregate accuracy and produced a sharp decline in Brown Spot F1. Moderated weighting produced a more favorable balance on the validation set. Importantly, the experiment does not establish square-root weighting as universally optimal; it only shows that this level of moderation was preferable among the three evaluated strategies on this dataset and architecture.

6.2 Leaf Smut remains a major weakness

Leaf Smut remained difficult even after weighting. The clean dataset count was 721 images compared with 2,238 Rice Tungro images and 2,176 Brown Spot images. The descriptive analysis found a positive association between class count and validation F1, with Spearman ρ equal to 0.543 p equal to 0.266 and sample size equal, to 6; this association is not statistically significant because the sample size of six classes is too small. Therefore, the study does not claim that class size caused the poor Leaf Smut performance. Other possible contributors include within-class visual variation, symptom similarity, disease severity, illumination, background, and annotation uncertainty.

6.3 Brown Spot and cross-class confusion

Brown Spot showed a particularly large validation-to-test decline and was frequently confused with Rice Tungro and Blast. This pattern suggests that visual similarity and distribution differences may be important. However, the present experiments do not include lesion-level annotations or controlled image acquisition, so causal interpretation should remain cautious.

6.4 Comparison with published literature

The final test accuracy is lower than many headline values reported in rice disease literature. This is not a contradiction in itself. Paddy Doctor, for example, reports 97.50% F1 for ResNet34 under its own dataset and benchmark protocol [5], while LiSA-MobileNetV2 reports 95.68% test accuracy in a different experimental setup [6]. These values are not directly comparable with the present results because the datasets, class definitions, sample sizes, acquisition conditions, architectures, preprocessing, and split protocols differ. The appropriate interpretation of this study is therefore reliability and generalization analysis rather than leaderboard ranking.

6.5 Implications for agricultural AI

The model performs substantially better on Healthy, Rice Tungro, and Sheath Blight than on Leaf Smut. This uneven reliability matters in agricultural decision support: a single aggregate accuracy number can conceal failure modes that are operationally important. A future deployment should therefore report class-wise sensitivity, calibrated confidence, and potentially an abstention mechanism rather than forcing a prediction for every image.

7. Limitations

First I notice that the study uses one public dataset that was collected in Bangladesh and as a result the study does not show how well the method works on Nepalese rice fields. Second I notice that although the 973-image test set is independent, from the training and validation subsets the test set still comes from the underlying dataset and therefore the study measures only within-dataset generalization instead of cross-dataset or cross-country generalization. Third I notice that the primary experiments use one lightweight architecture named MobileNetV2 and the study tests just three different weighting strategies. Fourth I notice that the exact-hash audit removes images that're pixel-identical but the study does not guarantee that there are no near-duplicates or correlated images taken during the same acquisition event. Finally, the Leaf Smut test subset contains only 72 images, which limits the precision with which class-specific performance can be estimated.

8. Future Work

The next stage should focus on external validation using independently collected field imagery, ideally including Nepalese rice varieties and multiple locations and seasons. Group-aware or session-aware splitting should be investigated to reduce correlation between images captured from the same plant or scene. Near-duplicate detection using perceptual hashes or feature embeddings could complement exact hashing. Methodologically, focal or balanced-focal loss, targeted augmentation, contrastive or metric learning, and calibration-aware objectives are promising directions. Model explainability, uncertainty estimation, and abstention should also be evaluated before any field deployment.

9. Conclusion

This study presented a leakage-resistant empirical evaluation of lightweight rice leaf disease classification under class imbalance. An exact-content audit reduced the original 9,769-image dataset to 9,722 unique images and eliminated all identified exact duplicate and conflicting-label groups. We compared three MobileNetV2 training strategies using the protocol. The MobileNetV2 training strategy that used class weighting gave the best validation results with an accuracy of 64.61 percent a Macro-F1 score of 58.41 percent and a Weighted-F1 score of 63.22 percent. When we tested the MobileNetV2 training strategies, on the held-out test set we got an accuracy of 60.12 percent and a Macro-F1 of 53.14 percent showing a clear generalization gap.

The main scientific lesson is not that moderated weighting solves rice disease classification. Rather, the results show that modest changes in imbalance handling can alter class-level behavior, and that held-out testing exposes weaknesses that aggregate validation metrics can obscure. Leaf Smut and Brown Spot were particularly challenging, while Sheath Blight, Rice Tungro, and Healthy were more reliably recognized. Future research should prioritize external field validation, robust split design, and uncertainty-aware evaluation before treating such models as dependable agricultural diagnostic systems.

10. Data and Reproducibility Statement

The study used the publicly released RiceLeafDiseaseBD Version 2 dataset. The dataset citation, version number, and DOI are specified in Reference [4]. The cleaned manifest, experimental metrics, confusion matrices, model checkpoints, and training history were organized in a research directory during experimentation. The held-out test set was locked before final evaluation. The reported results in this manuscript correspond to the cleaned 9,722-image dataset and the final 7,777/972/973 split.

11. References

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Author note. This manuscript intentionally avoids claims of state-of-the-art performance or production readiness. Its principal contribution is an empirical, leakage-aware evaluation emphasizing balanced metrics and held-out generalization.